

Security Architecture

Security architecture ensures that customer data and systems are protected through multiple layers of defense. It translates business and compliance requirements into executable security controls that safeguard the platform against potential threats. The architecture is designed to provide defense-in-depth by applying layered protection across the network, application, and database components of the platform.

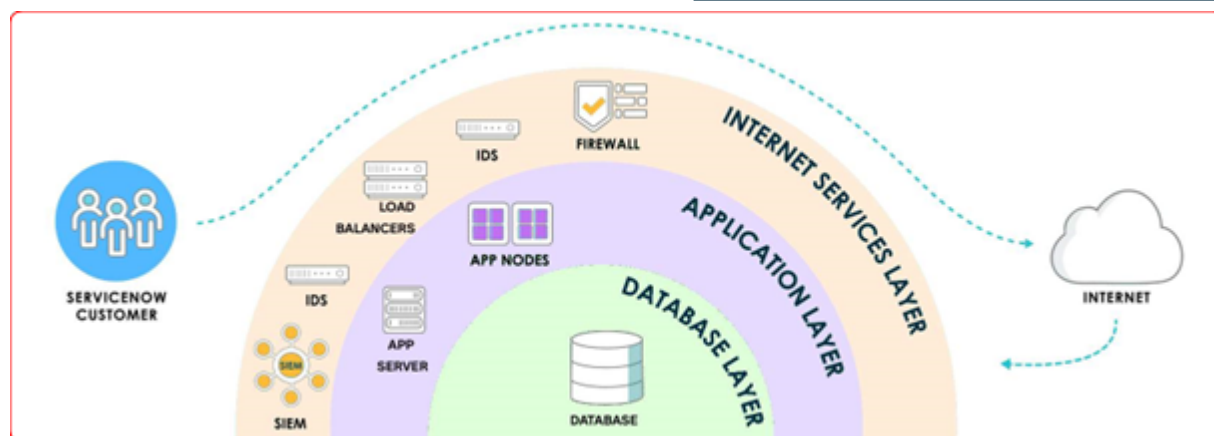
Security mindset questions and threat modeling

- A security mindset is the proactive approach of anticipating risks and identifying potential threats before they occur. It involves being skeptical, curious, detail-oriented, and preventive in thinking rather than reactive. When architecting a solution, apply a security mindset by asking yourself two key questions:
 - What might go wrong?
 - What can we do to prevent this?
- Threat modeling complements this mindset. It is a risk-based approach to designing secure systems that systematically identifies possible threats, assesses their impact, and defines mitigations. This process ensures that both technical and business stakeholders can anticipate and manage security risks effectively.

ServiceNow security architecture

There are two types of security architectures:

- **Physical security architecture:** The physical architecture of ServiceNow supports customers where they locate, operate, and store data globally. Its data centers are arranged in high-availability regional pairs across five continents: Asia, Australia, Europe, North America, and South America. All ServiceNow data centers uphold the highest security and reliability standards, including:
 - Physical access restricted to authorized ServiceNow personnel
 - 24/7 security guards and CCTV
 - Biometric access controls
 - Physical Intrusion detection and alarms
 - External anti-climb fencing and crash-resistant walls
 - Physical access audits
 - Multiple redundant power distribution paths
 - Environmental controls and fire detection and suppression systems
- **Logical security architecture:** There are three layers of logical security used in platform security architecture.
 - Network layer (Internet services layer)
 - Application layer
 - Database layer



The table below elaborates on the three layers of logical security.

Security layer	Description	Key security controls
Network layer	Focuses on securing network connectivity and data transmission.	• TLS 1.2 encryption: Encrypts all traffic to and from the instance. • IP address access control: Activates the IP Range Based Authentication plugin to define trusted and denied IP ranges, for instance access. • Certificates: Ensures that every customer instance establishes secure connections and validates signatures. • Virtual Private Network (VPN): Ensures that a user's IP is within the company's internal allowed network range.

Application layer	Focuses on secure user access, authentication, and instance configuration.	• Pre-logout: Enforces authentication controls to the right persona at the right time. • Adaptive authentication & Multi-Factor Authentication (MFA): Validates the identity of a user who accesses an instance. • Authorization & Role-Based Access Control (RBAC): • Allows role-based access and ACLs for data control. • Secures REST and Table APIs with ACLs, enforcing access controls and validations. • Zero Trust Access (ZTA): Enforces least-privilege access control, and operates on the “never trust, always verify” principle. • Security Center (SSC): Helps organizations maintain the security of their ServiceNow deployments.
Database layer	Protects stored data and the hardware in which the data is stored.	• Database encryption: Ensures that the data stored in the database remains secure and unreadable without decryption keys. • Platform encryption: Ensures ServiceNow data is encrypted at rest, allowing access only to authorized users. • ServiceNow Vault: Incorporates a set of security tools, such as code signing, data privacy, data discovery, and log export service, for protecting sensitive information.

Encryption types and description

The table below elaborates on the different types of encryptions and their features.

Encryption type	Description	Key features
Database encryption	Converts stored data into an unreadable format using a key and protects it from unauthorized access.	• Encrypts data stored in the database. • Makes data unusable without the decryption key. • Meets customer policies for data classification and compliance.
Platform encryption	Encrypts ServiceNow data at rest, ensuring only authorized users can access it.	Field encryption: • Encrypts sensitive data within supported field types permanently in the database and UI. • Uses Module Access Policies (MAPs) to restrict access for users, scripts, and system processes. • Includes access observer to track who/what accessed a field before encryption. Cloud encryption: • Encrypts total database host and backups for datacenter compliance. • Protects against physical theft or unauthorized access in datacenters.